CHAPTER 3

Section 3-1

3.1.7

A ∩ B = 70, A′ = 14, A ∪ B = 95

3.1.8

(a) $A' = \{x \mid x \geq 72.5\}$
(b) $B' = \{x \mid x \leq 52.5\}$
(c) $A \cap B = \{x \mid 52.5 < x < 72.5\}$
(d) $A \cup B = \{x \mid x > 0\}$

3.1.9 Consider the hospital emergency department data given below. Let A denote the event that a visit is to hospital 1, and let B denote the event that a visit results in admittance to any hospital.

	Hospital				
	1	2	3	4	Total
Total	5292	6991	5640	4329	22,252
LWBS	195	270	246	242	953
Admitted	1277	1558	666	984	4485
Not admitted	3820	5163	4728	3103	16,814

Determine the number of persons in each of the following events.

(a) $A \cap B$ (b) A' (c) $A \cup B$ (d) $A \cup B'$ (e) $A' \cap B'$

(a) $A \cap B = 1277$
(b) $A' = 22252 - 5292 = 16960$
(c) $A \cup B = 1685 + 3733 + 1403 + 2 + 14 + 29 + 46 + 3 = 6915$
(d) $A \cup B' = 195 + 270 + 246 + 242 + 3820 + 5163 + 4728 + 3103 + 1277 = 19044$
(e) $A' \cap B' = 270 + 246 + 242 + 5163 + 4728 + 3103 = 13752$

Section 3-2

3.2.3

From equation 3.1, the answer is 10! = 3,628,800

3.2.4

(a) From equation 3.4, the number of samples of size five is $\binom{140}{5} = \frac{140!}{5!135!} = 416{,}965{,}528$

(b) There are 10 ways of selecting one nonconforming chip and there are $\binom{130}{4} = \frac{130!}{4!126!} = 11{,}358{,}880$ ways of selecting four conforming chips. Therefore, the number of samples that contain exactly one nonconforming chip is $10 \times \binom{130}{4} = 113{,}588{,}800$

(c) The number of samples that contain at least one nonconforming chip is the total number of samples $\binom{140}{5}$ minus the number of samples that contain no nonconforming chips $\binom{130}{5}$. That is

$$\binom{140}{5} - \binom{130}{5} = \frac{140!}{5!135!} - \frac{130!}{5!125!} = 130{,}721{,}752$$

3.2.5

(a) Every arrangement selected from the 12 different components comprises a different design. Therefore, $12! = 479{,}001{,}600$ designs are possible.

(b) 7 components are the same, others are different, $\frac{12!}{7!1!1!1!1!1!} = 95040$ designs are possible.

(c) $\frac{12!}{3!4!} = 3326400$ designs are possible.

3.2.6

From the 5 defective parts, select 4, and the number of ways to complete this step is 5!/(4!1!) = 5
From the 45 non-defective parts, select 6, and the number of ways to complete this step is 45!/(6!39!) = 8,145,060
Therefore, the number of samples that contain exactly 4 defective parts is 5(8,145,060) = 40,725,300
Similarly, from the 5 defective parts, the number of ways to select 5 is 5!(5!1!) = 1
From the 45 non-defective parts, select 5, and the number of ways to complete this step is 45!/(5!40!) = 1,221,759
Therefore, the number of samples that contain exactly 5 defective parts is
1(1,221,759) = 1,221,759

Therefore, the number of samples that contain at least 4 defective parts is
40,725,300 + 1,221,759 = 41,947,059

Section 3-3

3.3.1

(a) S = {1, 2, 3, 4, 5, 6}
(b) 1/6
(c) 2/6
(d) 5/6

3.3.3

(a) P(A) = 30/100 = 0.30
(b) P(B) = 77/100 = 0.77
(c) P(A') = 1 – 0.30 = 0.70
(d) P(A∩B) = 22/100 = 0.22
(e) P(A∪B) = 85/100 = 0.85
(f) P(A'∪B) =92/100 = 0.92

3.3.5

(a) P(all hip surgeries before another type)= $\frac{\frac{8!}{3!5!}}{\frac{12!}{3!4!5!}} = \frac{8!4!}{12!} = \frac{1}{495} = 0.00202$

(b) P(begins with hip surgery)= $\frac{\frac{11!}{3!3!5!}}{\frac{12!}{3!4!5!}} = \frac{11!4!}{12!3!} = \frac{1}{3}$

(c) P(first and last are hip surgeries)= $\frac{\frac{10!}{2!3!5!}}{\frac{12!}{3!4!5!}} = \frac{1}{11}$

(d) P(first two are hip surgeries)= $\frac{\frac{10!}{2!3!5!}}{\frac{12!}{3!4!5!}} = \frac{1}{11}$

Section 3-4

3.4.1.

(a) $P(A \cup B \cup C)$ (b) $P(A \cap B \cap C)$ (c) $P(A \cap B)$ (d) $P[(A \cup B) \cap C]$ (e) $P(A' \cap B' \cap C')$

(a) P($A \cup B \cup C$) = P(A) + P(B) + P(C), because the events are mutually exclusive. Therefore,
P($A \cup B \cup C$) = 0.2+0.3+0.4 = 0.9
(b) P ($A \cap B \cap C$) = 0, because $A \cap B \cap C = \varnothing$
(c) P($A \cap B$) = 0 , because $A \cap B = \varnothing$
(d) P($(A \cup B) \cap C$) = 0, because $(A \cup B) \cap C = (A \cap C) \cup (B \cap C) = \varnothing$
(e) P($A' \cap B' \cap C'$) =1-[P(A) + P(B) + P(C)] = 1-(0.2+0.3+0.4) = 0.1

3.4.2

(a) P(High temperature and high conductivity)= 74/100 =0.74
(b) P(Low temperature or low conductivity)
= P(Low temperature) + P(Low conductivity) – P(Low temperature and low conductivity)
= (8+3)/100 + (15+3)/100 – 3/100
= 0.26
(c) No, they are not mutually exclusive. Because P(Low temperature) + P(Low conductivity)
= (8+3)/100 + (15+3)/100
= 0.29, which is not equal to P(Low temperature or low conductivity).

Section 3-5

3.5.1

(a) $P(A) = \frac{7+32}{100} = 0.39$

(b) $P(B) = \frac{13+7}{100} = 0.2$

(c) $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{7/100}{20/100} = 0.35$

(d) $P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{7/100}{39/100} = 0.1795$

3.5.3

(a) 4/499 = 0.0080
(b) (5/500)(4/499) = 0.000080
(c) (495/500)(494/499) = 0.98
(d) 3/498 = 0.0060
(e) 4/498 = 0.0080
(f) $\left(\frac{5}{500}\right)\left(\frac{4}{499}\right)\left(\frac{3}{498}\right) = 4.82 \times 10^{-7}$

3.5.5 Consider the hospital emergency room data given below. Let *A* denote the event that a visit is to hospital 4, and let *B* denote the event that a visit results in LWBS (at any hospital).

	Hospital				
	1	2	3	4	Total
Total	5292	6991	5640	4329	22,252
LWBS	195	270	246	242	953
Admitted	1277	1558	666	984	4485
Not admitted	3820	5163	4728	3103	16,814

Determine the following probabilities.

(a) $P(A \mid B)$ (b) $P(A' \mid B)$ (c) $P(A \mid B')$ (d) $P(B \mid A)$

(a) $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{242/22252}{953/22252} = \dfrac{242}{953} = 0.2539$

(b) $P(A' \mid B) = \dfrac{P(A' \cap B)}{P(B)} = \dfrac{(195+270+246)/22252}{953/22252} = \dfrac{711}{953} = 0.7461$

(c) $P(A \mid B') = \dfrac{P(A \cap B')}{P(B')} = \dfrac{(984+3103)/22252}{(22252-953)/22252} = \dfrac{4087}{21299} = 0.1919$

(d) $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{242/22252}{4329/22252} = \dfrac{242}{4329} = 0.0559$

Section 3-6

3.6.2

Let A and B denote the events that the first and second chips selected are defective, respectively.
(a) P(B) = P(B|A)P(A) + P(B|A')P(A') = (19/99)(20/100) + (20/99)(80/100) = 0.2
(b) Let C denote the event that the third chip selected is defective.

$$P(A \cap B \cap C) = P(C \mid A \cap B)P(A \cap B) = P(C \mid A \cap B)P(B \mid A)P(A)$$
$$= \frac{18}{98}\left(\frac{19}{99}\right)\left(\frac{20}{100}\right)$$
$$= 0.00705$$

3.6.3

(a) (0.88)(0.27) = 0.2376
(b) (0.12)(0.13+0.52) = 0.0.078

3.6.4 Consider the hospital emergency room data given below. Let *A* denote the event that a visit is to hospital 4 and let *B* denote the event that a visit results in LWBS (at any hospital).

	Hospital				
	1	2	3	4	Total
Total	5292	6991	5640	4329	22,252
LWBS	195	270	246	242	953
Admitted	1277	1558	666	984	4485
Not admitted	3820	5163	4728	3103	16,814

Determine the following probabilities.

(a) $P(A \cap B)$ (b) $P(A \cup B)$ (c) $P(A' \cup B')$ (d)Use the total probability rule to determine $P(A)$

P(A) = 4329/22252 = 0.1945, P(B) = 953/22252 = 0.0428
(a) P(A ∩ B) = P(A | B)P(B) = (242/953)(953/22252) = 0.0109
(b) P(A ∪ B) = P(A) + P(B) – P(A ∩ B) = 0.1945 + 0.0428 – 0.0109 = 0.2264
(c) P(A′ ∪ B′) = 1 – P(A ∩ B) = 1 – 0.0109 = 0.9891
(d) P(A) = P(A | B)P(B) + P(A | B')P(B') = (242/953)(953/22252) + (4087/21299)(21299/22252) = 0.1945

Section 3-7

3.7.2

(a) $P(B \mid A) = 4/499$ and
$P(B) = P(B \mid A)P(A) + P(B \mid A')P(A') = (4/499)(5/500) + (5/499)(495/500) = 5/500$
Therefore, *A* and *B* are <u>not</u> independent.
(b) *A* and *B* are independent.

3.7.4

Let A denote the upper devices function. Let B denote the lower devices function.
P(A) = (0.9)(0.8)(0.7) = 0.504
P(B) = (0.95)(0.95)(0.95) = 0.8574
P(A∩B) = (0.504)(0.8574) = 0.4321
Therefore, the probability that the circuit operates = P(A∪B) = P(A) +P(B) – P(A∩B) = 0.9293

3.7.5

P = [1 – (0.1)(0.05)][1 – (0.1)(0.05)][1 – (0.2)(0.1)] = 0.9702

3.7.6 Consider the hospital emergency room data given below. Let A denote the event that a visit is to hospital 4, and let B denote the event that a visit results in LWBS (at any hospital). Are these events independent?

	Hospital				
	1	2	3	4	Total
Total	5292	6991	5640	4329	22,252
LWBS	195	270	246	242	953
Admitted	1277	1558	666	984	4485
Not admitted	3820	5163	4728	3103	16,814

P(A) = 4329/22252 = 0.1945, P(B) = 953/22252 = 0.0428, P(A ∩ B) = 242/22252 = 0.0109
Because P(A)*P(B) = (0.1945)(0.0428) = 0.0083 ≠ 0.0109 = P(A ∩ B), A and B are not independent.

Section 3-8

3.8.1

Let G denote a product that received a good review. Let H, M, and P denote products that were high, moderate, and poor performers, respectively.

(a)

$$\begin{aligned} P(G) &= P(G|H)P(H) + P(G|M)P(M) + P(G|P)P(P) \\ &= 0.95(0.40) + 0.60(0.35) + 0.10(0.25) \\ &= 0.615 \end{aligned}$$

(b) Using the result from part (a)

$$P(H|G) = \frac{P(G|H)P(H)}{P(G)} = \frac{0.95(0.40)}{0.615} = 0.618$$

© $$P(H|G') = \frac{P(G'|H)P(H)}{P(G')} = \frac{0.05(0.40)}{1-0.615} = 0.052$$

3.8.3

Let F denote a fraudulent user and let T denote a user that originates calls from two or more metropolitan areas in a day. Then,

$$P(F|T) = \frac{P(T|F)P(F)}{P(T|F)P(F) + P(T|F')P(F')} = \frac{0.30(0.0001)}{0.30(0.0001) + 0.01(.9999)} = 0.003$$

$$\begin{aligned} P(D \mid L) &= \frac{P(L \mid D)P(D)}{P(L \mid A)P(A) + P(L \mid B)P(B) + P(L \mid C)P(C) + P(L \mid D)P(D)} \\ &= \frac{(0.0559)(0.1945)}{(0.0368)(0.2378) + (0.0386)(0.3142) + (0.0436)(0.2535) + (0.0559)(0.1945)} \\ &= 0.2540 \end{aligned}$$

3.8.5

F: Free; S: Spam; V: Valid

P(F|S) = 0.6, P(F|V) = 0.04

(a) P(F) = P(F|S) P(S)+P(F|V) P(V) = 0.6(0.2) + 0.04(0.8) = 0.152

(b) P(S|F) = $\frac{P(F|S)P(S)}{P(F)} = \frac{0.6(0.2)}{0.152} = 0.789$

(c) P(V|F') = $\frac{P(F'|V)P(V)}{P(F')} = \frac{(0.96)0.8}{1-0.152} = 0.906$